

INVERTEBRATES

With about 1.4 million species identified, animals make up the largest kingdom of living things on Earth. The vast majority of them are invertebrates – animals without a backbone. Invertebrates are extraordinarily varied; many are microscopic, but the largest are over 10 m (33 ft) long.

Invertebrates were the first animals to evolve. Initially, they were small, soft-bodied, and aquatic – features that many living invertebrates still share. During the Cambrian Period, which ended about 485 million years ago, invertebrates underwent a spectacular burst of evolution, developing a huge range of body forms and some very different ways of life. This evolutionary explosion produced almost all the major phyla, or groups, of invertebrates that exist today.

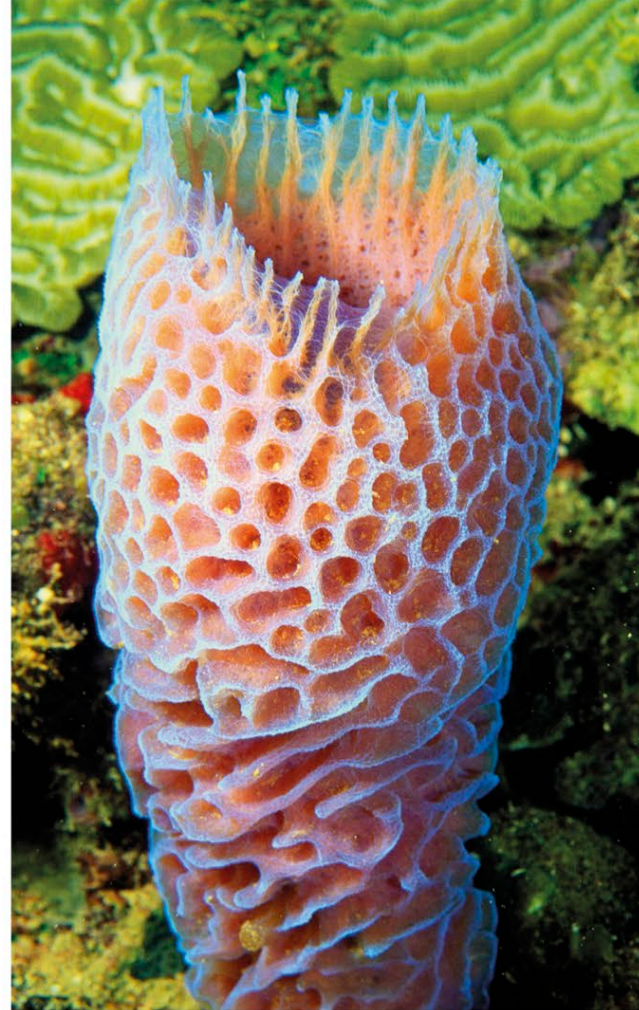
GREAT DIVERSITY

There is no such thing as a typical invertebrate, and many of the phyla have very little in common. The simplest kinds do not have a head or a brain, and usually rely on internal fluid pressure to keep their shape. At the other extreme, arthropods have a well-developed nervous system and elaborate sense organs, including complex eyes. Crucially, they also have an external body case, or exoskeleton, with legs that bend at flexible joints. This particular body plan has proved

to be remarkably successful, allowing arthropods to invade every natural habitat in water, on land, and in the air. Invertebrates also include animals that have a shell, and ones that are reinforced by mineral crystals or hard plates. Only vertebrates, however, have an internal skeleton made of bone.

SPLIT LIVES

Most invertebrates begin life as an egg. Some look like miniature versions of their parents when they hatch, but many start life with a very different body form. These larvae change shape, food sources, and feeding methods as they grow up. For example, sea urchins have drifting larvae, which filter food from the sea, while the adults scrape algae from rocks. The change in shape – metamorphosis – can be gradual, or it can take place abruptly, with the young animal's body being broken down, and an adult body being assembled in its place. Metamorphosis enables invertebrates to exploit more than one food source, and it also helps them to spread, often over great distances.



SPONGES

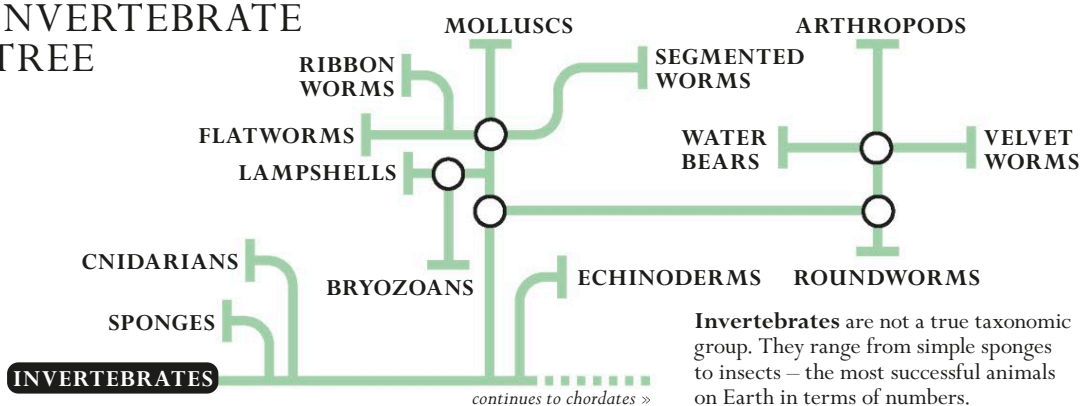
Among the simplest animals, sponges have a sieve-like body, with an internal skeleton of mineral crystals. Classified as the phylum Porifera, there are more than 9,000 species.

ARTHROPODS

The phylum Arthropoda is the biggest in the animal kingdom, with over 1.2 million species identified. It includes insects, crustaceans, arachnids, centipedes, and millipedes.



INVERTEBRATE TREE



Invertebrates are not a true taxonomic group. They range from simple sponges to insects – the most successful animals on Earth in terms of numbers.